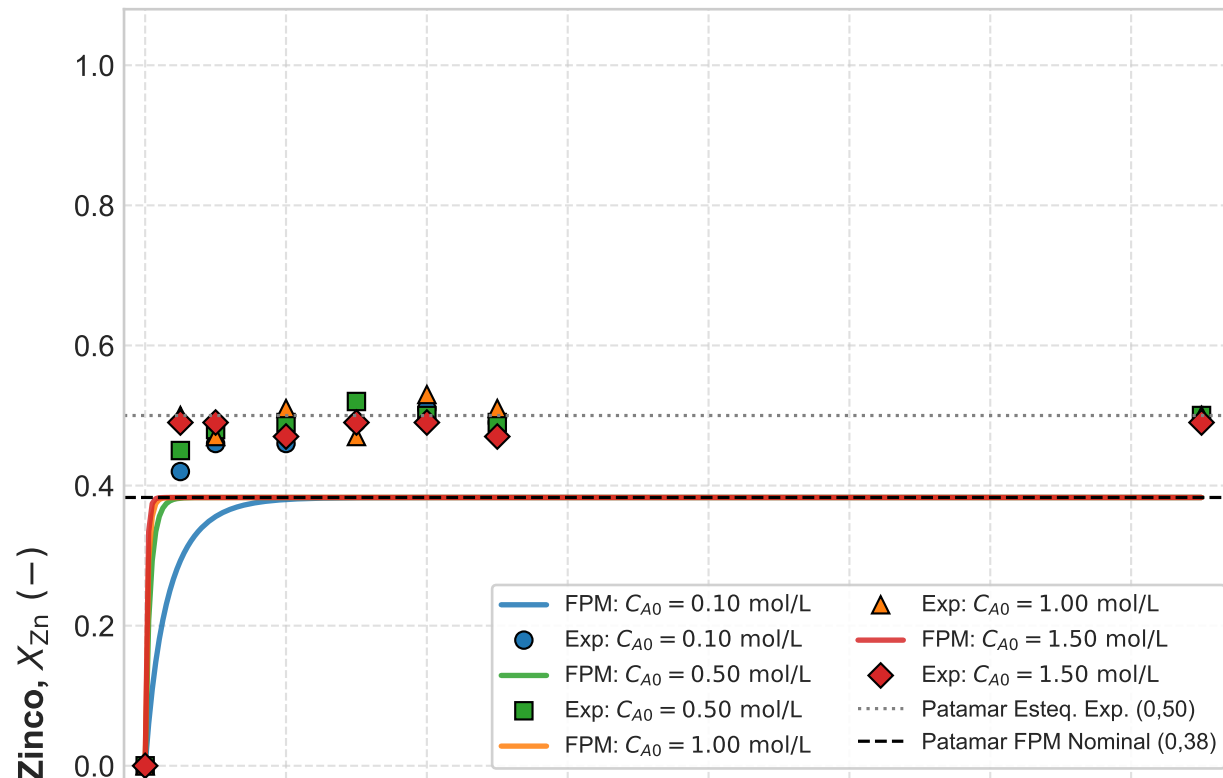
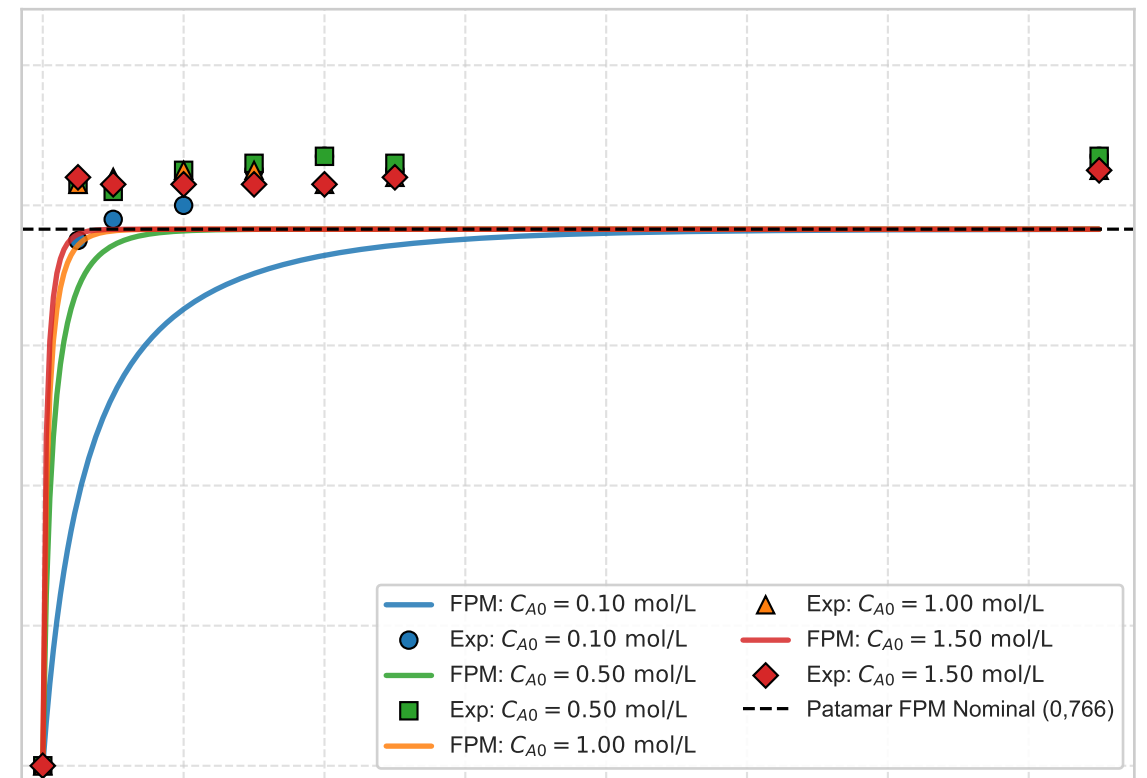


Simulação do Baseline Fenomenológico Puro (FPM Puro) vs. Dados Experimentais de Bancada
Modelo PBM com $\alpha_{\text{nominal}} = 5500 \mu\text{m}/\text{min}$ | Métricas Globais: $R^2 = 0.9030$, RMSE = 0.1010, MAE = 0.0699

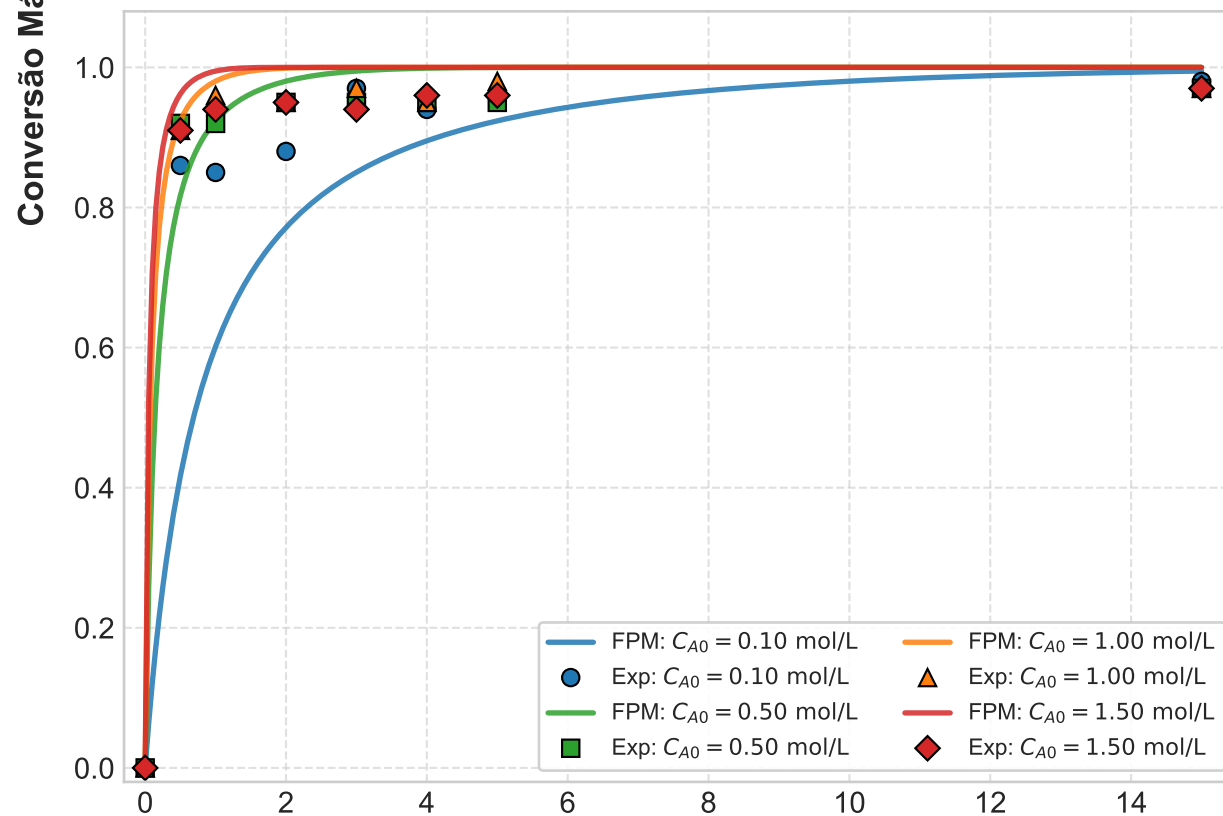
Painel (a) — Razão Molar $\eta = 0.5$



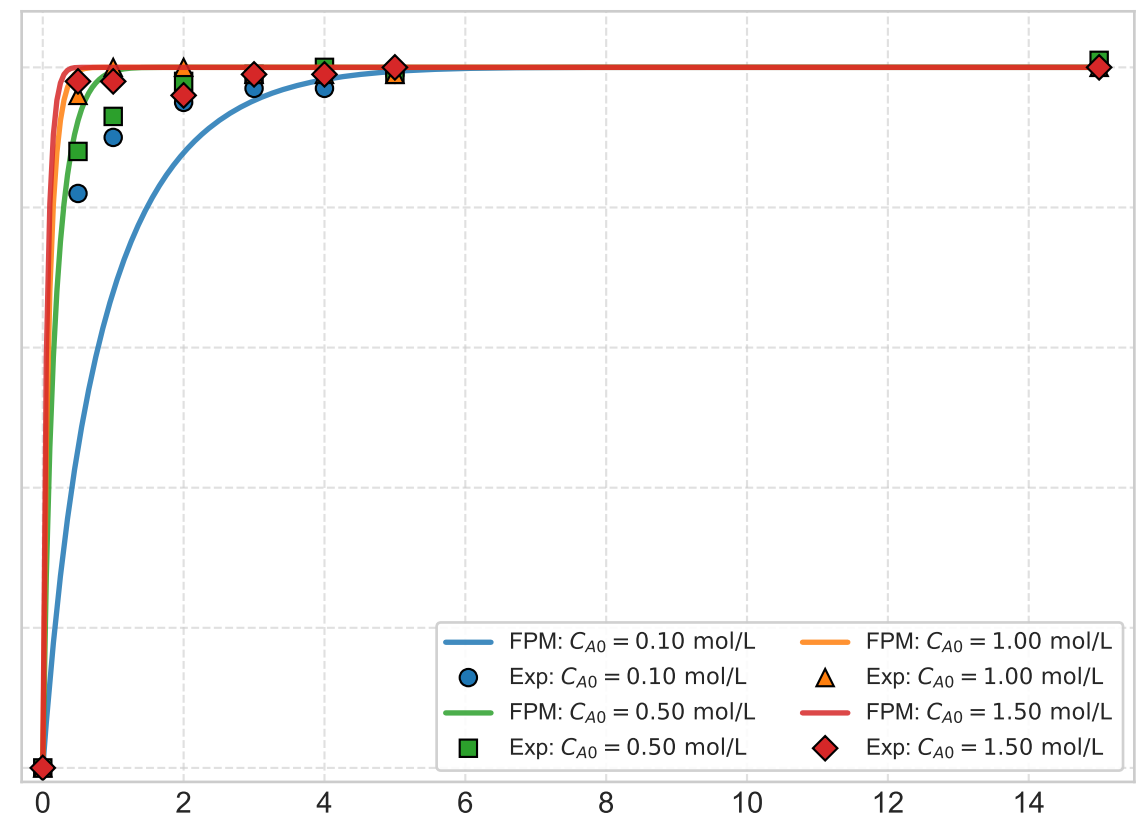
Painel (b) — Razão Molar $\eta = 1.0$



Painel (c) — Razão Molar $\eta = 1.5$



Painel (d) — Razão Molar $\eta = 3.1$



Tempo de Reação, t (min)